

Solution QUIZ 2 (Dated: 23 Oct 2024)
Data Structures and Algorithm Analysis

Roll No: _____

Name: _____

Q1. Write a member function for class Linked List to print all pair of nodes, which are in order (left element is smaller or equal to right element)?

```
for (Node *t = head ; t -> next ; t = t -> next)
    if (t -> val <= t -> next -> val)
        cout << t -> val << ' ' << t -> next -> val << '\n';
```

Q2. Consider linked list with three pointers pointing at particular nodes. Perform next operations. Redraw list & write first six outputs of the print statement. Ignore any logical issues?

head -> 23 -> 46 -> 39 -> 24 -> 48 -> 35 -> x



t1
t2
t3

t1->next=t3;

t2->next=t1;

head->next=t3;

t3->next=t2;

```
for (t1 = head ; t1 != NULL ; t1 = t1 -> next)
```

```
    cout << t1 -> data << ' ';
```

Q3. Write a class **Record** having class record. The class will store, roll nos and cgpa's of the students. Write following member functions:

- | | |
|---|--|
| - int getCount() const | - int indexMaxCGPA() const |
| - void addRecord(int roll_no, float cgpa) | - void modifyRecord(int roll_no, float cgpa) |

```
class Record{
    vector<int> rnos;
    vector<float> cgpas;
public:
    int getCount() const{
        return rnos.size();
    }
    void addRecord(int roll_no, float cgpa){
        rnos.push_back(roll_no);
        cgpas.push_back(cgpa);
    }
    int indexMaxCGPA() const{
        int i, maxI = 0;
        for (i = 1 ; i < cgpas.size() ; i++)
            if (cgpas[maxI] < cgpas[i]) maxI = i;
        return maxI;
    }
    void modifyRecord(int roll_no, float cgpa){
        for (int i = 0 ; i < cgpas.size() ; i++)
            if (rnos[i] == roll_no){
                cgpas[i] = cgpa;
                break;
            }
    }
};
```